DATA VISUALISATION

What is Data Visualisation?

Data visualization is the **representation of data** or information in a **graph, chart, or other visual format.**

It **communicates relationships of the data with images.**

Why Data Visualisation?

A visual summary of information makes it **easier to identify patterns, trends and outliers** than looking through **thousands of rows on a spreadsheet**.

It's the way the human brain works.

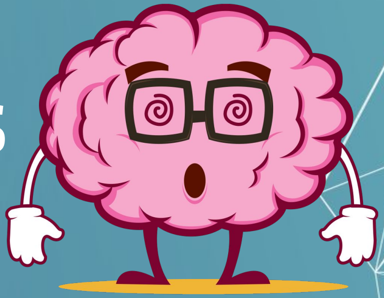
Since the purpose of data analysis is to gain insights, **data is much more valuable when it is visualized.**

"A picture is worth a thousand words"

Our **eyes are drawn to colors and patterns.** We can quickly identify red from blue, square from circle. Our culture is visual, including everything from art and advertisements to TV and movies

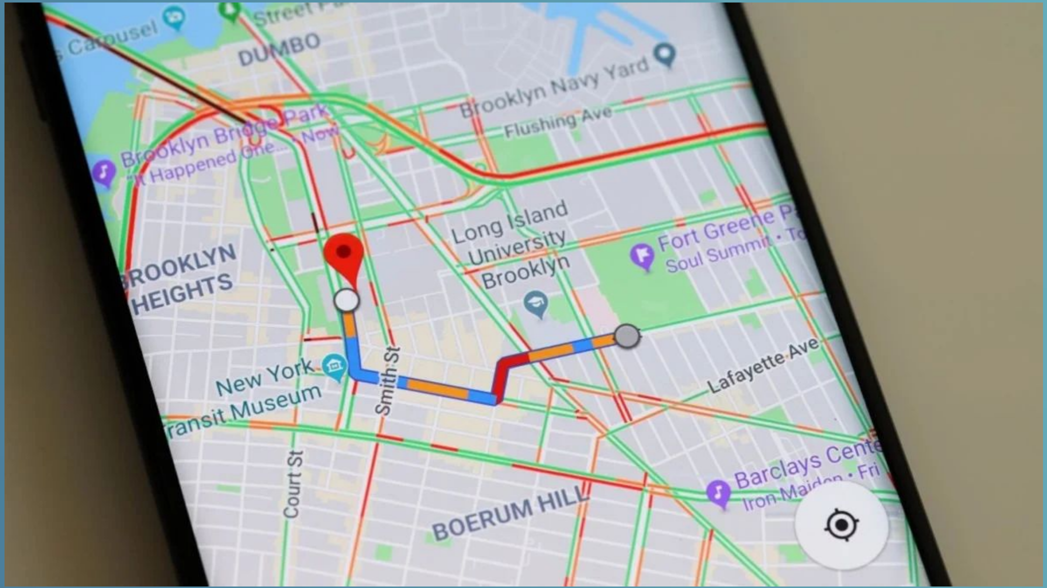
Data visualization is another form of visual art that **grabs our interest and keeps our eyes on the message.**

You’ve been surrounded by Data Visualisations all this while!



Google Map Traffic visualization

The live traffic status in Google map is another simple example of what is possible through data visualization to make humans life simple and easy. You can see the **red colour indicating slow traffic area** in the city on the map and **blue colour indicating fast** etc.

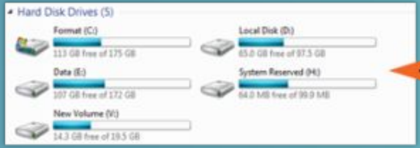


Hard Disk Drives interactive Data Visualization

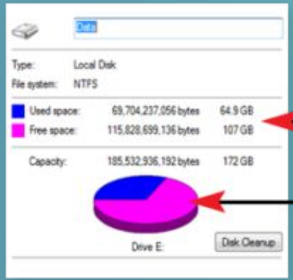
The **blue line and empty space in horizontal bar** and in **pie chart** is an example to **inform about the consumed and free space in hard disk.**

Hard drives data status in interactive visualization is **effective to understand even for a normal user.**

And that's always the target of data visualization to make it so much easy and relevant for people to understand and take quick decision.



Hard disk data status
in interactive visualization



Hard disk data status
in interactive Pie Chart visualization

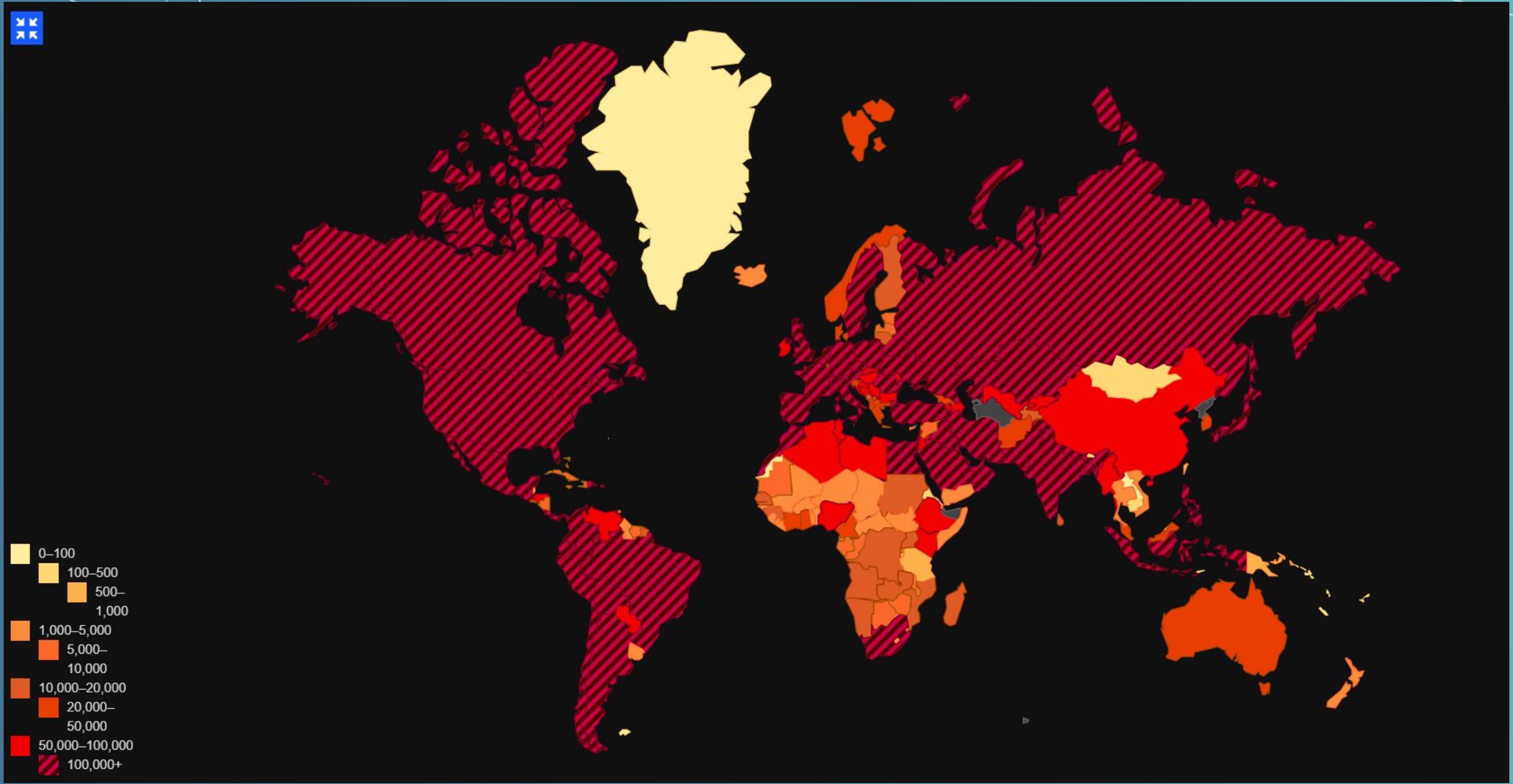
Geo Spatial Data Visualization

Geospatial visualizations are one of the earliest forms of information visualizations.

Amongst the various types of geospatial visualisations, **Heat maps** are useful when you have to represent large sets of continuous data on a map using a color spectrum.

The map of the world in the next slide shows the world population affected with covid using different shades of red. **The darker the shade of red, the higher the affected population.**

Geo Spatial Data Visualization



Source: wuflu.live

Can you think of more such examples?

Data Visualisation in Python

Python offers **multiple great graphing libraries** that come packed with **lots of different features**. No matter if you want to **create interactive, live or highly customized plots,** python has an excellent library for you.

To get a little overview here are **a few popular plotting libraries:**

- **Matplotlib**
- **Pandas Visualization**
- **Seaborn**
- **ggplot**
- **Plotly**
- **Bokeh**
- **Kepler.gl**

Matplotlib

Matplotlib is the **most popular python plotting library.**
It is a **low-level library** with a Matlab like interface which offers lots of freedom at the cost of having to write more code.

Seaborn

Seaborn is a **graphic library built on top of Matplotlib.**

It allows to **make your charts prettier,** and facilitates some of the common data visualisation needs.

Let’s see how to work with these two data visualisation libraries

Dataset we'll be working on

Standard_Metropolitan_Areas_Data

It contains **data of 99 standard metropolitan areas in the US**. The data set provides information on **10 variables** for each area for the period 1976-1977. The areas have been divided into 4 geographic regions: 1=North-East, 2=North-Central, 3=South, 4=West. The variables provided are listed in the table below:

Variable name	Description
land_area	size in square miles
total_population	estimated population in thousands
percent_city	percent of population in central city/cities
percent_senior	percent of population ≤ 65 years
physicians	number of professionally active physicians
hospital_beds	total number of hospital beds
graduates	percent of adults that finished high school
work_force	number of persons in work force in thousands
income	total income in 1976 in millions of dollars
crime_rate	Ratio of number of serious crimes by total population
region	geographic region according to US Census

Slide Download Link

You can download these slides from the below link:

**https://docs.google.com/presentation/d/1Khjv0m9A8hUTKM9ZS-iPFFLtHILVVS
MIOSb8fzavISs/edit?usp=sharing**

That's it for this session

Thanks!

QnA Time